

A Subcooled Pool-Boiling Chamber with a Submerged Tube-Bundle Condenser: A Validated Reduced-Order Predictive Thermal Model

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Abstract

This paper presents a validated reduced-order predictive thermal model of a closed, subcooled pool-boiling chamber in which a heated chip boils a dielectric working fluid and the vapor is rejected to coolant through a partially submerged tube-bundle condenser, a configuration developed for high-heat-flux data-center cooling. The model is a mechanistically grounded resistance network with a small set of physically interpretable constants: a transferable surface-fluid coefficient for nucleate boiling, a fixed area augmentation for the open-microchannel surface ($A_r = 2.17$), and a compact operating-point map for the coolant-side-limited condenser. It predicts chip surface temperature versus heat flux across two evaporator surfaces (plain copper and open microchannel), two condenser bundles (33-tube and 42-tube), and four coolant inlet temperatures, reproducing 261 experimental points with a root-mean-square error of 4.4 K. The plain- and microchannel-surface coefficients ($C_{sf} = 0.0131$ and 0.0067) recover independently on geometrically different chambers and transfer between them, supporting their use as surface-fluid properties rather than per-chamber fits. Out-of-sample tests, leaving out one coolant inlet and transferring the surface coefficient across chambers, support the predictive claim and localize the residual to the operating-point representation of the confined-bundle condenser. The calibrated model is then used to predict the performance of a denser, untested 66-tube condenser, whose lower coolant-side resistance follows from its smaller tube bore.

1 Introduction

Two-phase immersion cooling of high-power electronics relies on nucleate boiling at a heated surface and condensation of the generated vapor at a heat exchanger. In a closed chamber the two processes are coupled: the chamber pressure, hence the saturation temperature T_{sat} at which the fluid boils and condenses, settles so that the condenser rejects exactly the heat the chip dissipates. Predicting chip temperature for an untested geometry therefore requires not only a boiling correlation but also a prediction of the operating point itself. The objective is a genuinely predictive model that forecasts chip surface temperature for new condenser and evaporator geometries, coolant inlet temperatures, and chip surfaces, rather than curve-fitting a single data set. The model is a reduced-order “gray box”: mechanistic correlations carry the physics, and a minimal number of physically interpretable calibrated constants absorb what cannot be resolved from geometry alone.

2 Literature review

Thermal management context. The rising power density of microprocessors and the growth of data-center workloads have made conventional air cooling insufficient, motivating direct liquid and two-phase schemes [1, 2]. Two-phase immersion cooling, in which a chip is immersed in a dielectric or water bath that boils on contact and condenses elsewhere in the enclosure, offers very high heat transfer coefficients and near-isothermal operation [3, 4]. The present chamber is a closed realization of this concept with a submerged condenser, so its performance depends jointly on the boiling surface and the condenser.

The present apparatus. The chamber modeled here was introduced by Shukla and Kandlikar [6], who combined subcooled pool boiling with a partially submerged tube-bundle condenser in a compact form factor that fits a 1U/2U server rack. Partly submerging the condenser augments subcooled boiling and adds direct-contact condensation, and high-speed imaging confirmed submerged condensation with small, discrete vapor bubbles. The reported thermal test vehicle dissipated about 400 W at a surface temperature below 80 °C on Novec-7000, and over 1 kW from a 34.5×32 mm plain copper chip on water, establishing the operating envelope this model is built to predict.

Pool boiling and the boiling curve. The characteristic boiling curve, with its single-phase, nucleate, transition and film regimes, was established by Nukiyama [8] and treated comprehensively in the monographs of Carey [13] and Incropera et al. [14]. The most widely used nucleate-boiling correlation is that of Rohsenow [9], in which a surface–fluid coefficient C_{sf} encapsulates the combined effect of surface finish, wettability and fluid. Mikic and Rohsenow [11] refined the treatment of surface characteristics, and Piro [12] compiled experimental C_{sf} values, emphasizing that the coefficient is a property of the specific surface–fluid pair and must be calibrated. This motivates fitting C_{sf} once per chip surface and treating it as a transferable constant.

Critical heat flux. The peak nucleate flux, or critical heat flux (CHF), was first described by hydrodynamic instability theory [15, 16], later extended to finite and oriented surfaces by Lienhard and Dhir [17]. Kandlikar [18] introduced a force-balance model incorporating the dynamic receding contact angle and surface orientation; Mudawar and co-workers developed complementary near-saturated CHF models [19]. Because no burnout data beyond the measured CHF points were available, CHF is anchored to measurement rather than modeled mechanistically.

Enhanced surfaces and open microchannels. Surface structuring is the principal route to higher CHF and heat transfer coefficient. Cooke and Kandlikar [20, 21] showed that open microchannels milled into copper dissipate fluxes on the order of 240 W/cm² at low wall superheat, and identified channel rewetting and separate liquid–vapor pathways, rather than mere area increase, as the dominant mechanism. Subsequent work combined microchannels with sintered or electrodeposited porous coatings [24, 25] and demonstrated bubble-induced convective liquid feeding [26]. The chip modeled here uses the Cooke–Kandlikar geometry, represented by a fixed geometric area augmentation with a reduced C_{sf} .

Dielectric working fluids. Immersion cooling favors dielectric coolants, fluorocarbons and hydrofluoroethers such as Novec/HFE-7000, for their electrical inertness and low, near-ambient saturation temperature (about 34 °C for HFE-7000 at atmospheric pressure), at the cost of a much

lower critical heat flux than water. El-Genk and Pourghasemi [22, 23] measured saturated and subcooled boiling of HFE-7000 on copper of varying roughness and orientation, correlating both the heat transfer coefficient and CHF against surface roughness, subcooling, and inclination; they report CHF on plain copper near 20–30 W/cm², rising 50–70% at 30 K subcooling. These dependences, roughness, subcooling, and contact angle, are the levers the present CHF closure must capture when the model is extended from water to HFE-7000.

Condensation. The condenser side rests on the classical film-condensation analysis of Nusselt [27], extended to dropwise condensation [28] and to tube bundles in the standard references [29]. A central finding here is that, for this geometry, condensation is not the limiting resistance.

Internal convection. The coolant-side resistance is governed by internal forced convection. For the short, small-bore tubes used here the flow is laminar and thermally developing, a regime described by Hausen’s entrance-region correlation [30] and tabulated by Shah and London [31]; the turbulent limit is covered by Gnielinski [32], and external natural convection by Churchill and Chu [33]. Recognizing the developing-laminar character of the coolant flow is what corrects an earlier over-prediction of the operating point.

System-level studies and reduced-order models. At the system level, immersion cooling is pursued in both single-phase [7] and two-phase [4] forms, and recent multi-chip and loop studies emphasize that overall performance is set as much by the condenser and the coolant loop as by the boiling surface. Detailed CFD of such chambers is costly, which has motivated semi-empirical and lumped models for design-space exploration; the gray-box developed here follows that line, pairing mechanistic correlations with a small set of calibrated, physically interpretable constants rather than a full numerical solution.

Closed-chamber coupling and the gap addressed here. Closed two-phase enclosures and thermosyphons couple evaporation and condensation through the vapor space [34], and non-condensable gases and headspace pressure shift the operating point. Yet most boiling and condensation studies treat the two processes in isolation, and few predict the coupled operating point of a closed chamber from geometry alone. The present model addresses that gap with a calibrated, mechanistically grounded operating-point sub-model.

3 Methods

3.1 Chip and condenser geometry

Both chips share an 11.04 cm² (34.5 × 32 mm) copper footprint and are mounted at the base of the chamber. The plain chip is polished to $R_a \approx 2.5 \mu\text{m}$. The enhanced chip carries open microchannels of the Cooke–Kandlikar type [21]: rectangular channels 381 μm wide and 400 μm deep separated by 250 μm fins, increasing the wetted area and providing rewetting pathways. Three submerged copper condensers are considered (Table 1). The 33-tube design uses relatively large 4.76 mm tubes with external (open) manifolds; the 42-tube design uses denser, smaller 3.175 mm tubes with in-chamber manifolds, placing a lower, more submerged bank in the pool; the 66-tube design exists

in CAD but has no experimental data. The coolant is water, supplied at ~ 78 g/s (33-tube) and ~ 36 g/s (42-tube).

Table 1: Condenser and chip geometry.

Condenser	Tubes N	OD (mm)	ID (mm)	L (mm)	Ext. area (cm ²)
33-tube (open manifolds)	33	4.76	3.14	95.0	469
42-tube (in-chamber manifolds)	42	3.175	1.39	62.6	262
66-tube (no data yet)	66	2.00	1.80	95.0	394
<i>Chip surfaces</i>					
Plain	polished copper, $R_a \approx 2.5$ μm				
Microchannel	open channels 381 μm wide $\times$ 400 μm deep, 250 μm fins				

The two condenser bundles are shown in cross-section in Figure 1, reconstructed from the CAD. The 33-tube bundle is a single bank of 33 relatively large straight tubes in five staggered rows; the 42-tube bundle is denser, with 42 smaller-diameter straight tubes in four staggered rows fed by in-chamber manifolds. All tube axes are parallel; the tubes are straight, not U-shaped.

Condenser tube layouts (from CAD)

Copper tube walls (amber) and coolant bores (blue), drawn to common scale.

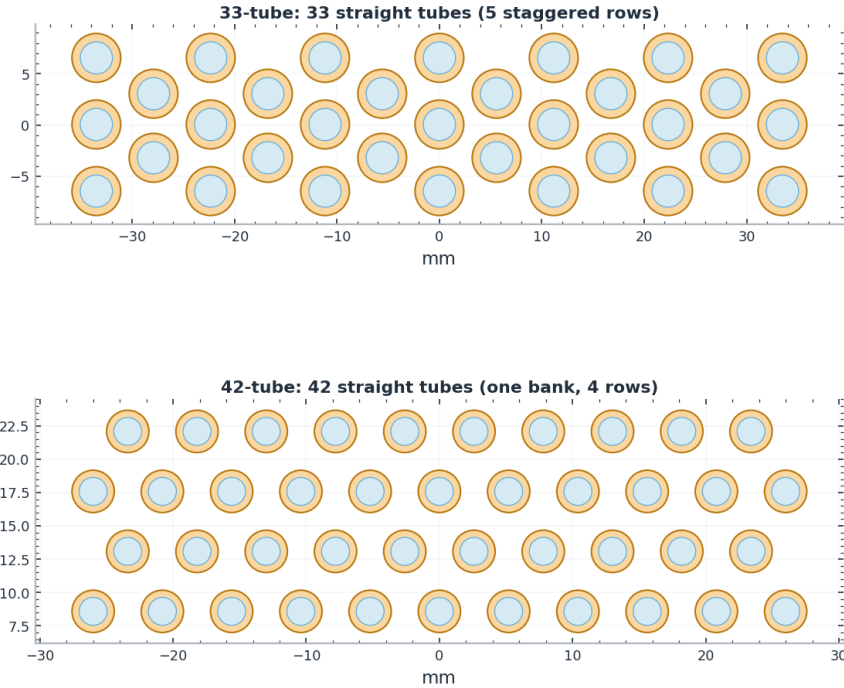


Figure 1: Condenser cross-sections drawn to a common scale from the CAD geometry, looking along the tube axis (coolant into the page). The 33-tube bundle uses larger straight tubes at wider pitch; the 42-tube bundle uses more, smaller straight tubes at tighter pitch.

3.2 Modeling domain

The model resolves the chip-to-coolant path as the reduced thermal network of Figure 2. Three temperatures are tracked: the chip surface T_{surf} , the bulk (pool) liquid T_{bulk} , and the coolant inlet T_{in} . Heat Q flows from the chip into the pool across a boiling/chip resistance and from the pool to the coolant across a condenser resistance, the two acting in series. The pool is treated as a single well-mixed reservoir at $T_{bulk} = T_{sat} - \Delta T_{sub}$; the vapor space and chamber pressure are represented implicitly through the saturation temperature; and the condenser is reduced to a coolant-side-limited heat exchanger characterized by the operating-point map. The chip surface is a lumped boiling node rather than a spatially resolved field. This reduction is justified a posteriori by the resistance breakdown, which shows the path is dominated by two clearly separated resistances.

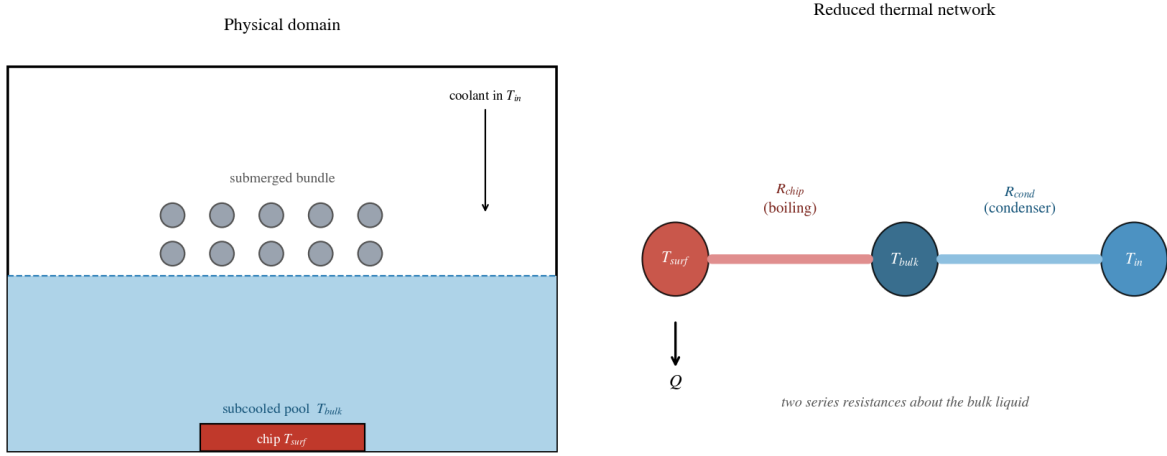


Figure 2: Physical domain (left) and the reduced thermal network (right). The chip boils the subcooled pool; vapor condenses on the submerged bundle and rejects heat to the internal water coolant. The chip-to-coolant path reduces to two series resistances about the bulk liquid temperature.

3.3 Governing equations

The chip dissipates heat by simultaneous single-phase convection into the subcooled pool and nucleate boiling. The nucleate component follows the Rohsenow correlation [9],

$$q''_{nb}(T_s) = \mu_l h_{fg} \sqrt{\frac{g(\rho_l - \rho_v)}{\sigma}} \left[\frac{c_{p,l}(T_s - T_{sat})}{C_{sf} h_{fg} Pr_l^s} \right]^3, \quad T_s > T_{sat}, \quad (1)$$

with $s = 1$ for water. The single-phase component is a horizontal-surface natural-convection coefficient scaled by a per-chamber factor F_{nc} ,

$$q''_{nc}(T_s) = F_{nc} h_{nc}(T_s, T_{bulk}) (T_s - T_{bulk}). \quad (2)$$

The mechanisms combine in quadrature, and the microchannel surface uses the same expressions multiplied by a geometric area-augmentation ratio A_r :

$$q'' = A_r \sqrt{q''_{nc}(T_s)^2 + q''_{nb}(T_s)^2}. \quad (3)$$

The microchannel critical heat flux exceeds the plain value by a near-constant, geometry-independent increment,

$$q''_{CHF,micro} = q''_{CHF,plain} + \Delta q''_{CHF}, \quad \Delta q''_{CHF} \approx 60 \text{ W/cm}^2. \quad (4)$$

The chamber operating point closes the system. A resistance breakdown (Section 4) shows the condenser is coolant-side limited, with internal flow developing-laminar ($Re \sim 1100$) and film condensation nearly ideal and almost negligible; the coolant-side conductance improves as the system warms, so T_{sat} responds to coolant inlet with a gain below unity. The developing-laminar Nusselt number [30],

$$Nu = 3.66 + \frac{0.0668 Gz}{1 + 0.04 Gz^{2/3}}, \quad Gz = \frac{D_i}{L} Re Pr, \quad (5)$$

reproduces the magnitude and temperature dependence of the dominant resistance and motivates a calibrated linear operating-point map per chamber,

$$T_{sat} = a T_{in} + b Q + c, \quad (6)$$

$$\Delta T_{sub} = \max(d T_{in} + e Q + f, 0), \quad (7)$$

$$Q = q'' A_{foot}, \quad A_{foot} = 11.04 \text{ cm}^2.$$

The chip-to-coolant path is split at the bulk liquid into two series resistances,

$$R_{chip} = \frac{T_s - T_{bulk}}{Q}, \quad R_{cond} = \frac{T_{bulk} - T_{in}}{Q}, \quad R_{tot} = R_{chip} + R_{cond} = \frac{T_s - T_{in}}{Q}. \quad (8)$$

3.4 Solver setup

The model is implemented in Python (NumPy/SciPy). Thermophysical properties are evaluated with CoolProp for water (**Water**). HFE-7000 is not available in the CoolProp build used here, so its properties are taken from the literature [22] and held fixed ($h_{fg} = 142 \text{ kJ/kg}$, $\sigma = 12.4 \text{ mN/m}$, $T_{sat} \approx 34 \text{ }^\circ\text{C}$ at 1 atm). Properties are evaluated at the saturation or film temperature and cached. The fully predictive chain proceeds as follows: from the inputs (q'' , T_{in} , geometry) the operating-point map, Eqs. (6)–(7), returns T_{sat} and ΔT_{sub} , hence T_{bulk} ; the chip relation, Eq. (3), is then solved for the surface temperature T_s by Brent’s method on the bracket $[T_{bulk}, T_{sat} + 260 \text{ K}]$, widened to +1000 K (water) or +200 K (HFE-7000) if the root is not initially bracketed, to a tolerance of 10^{-6} K . No measured operating condition enters the prediction. Boiling curves and parity plots are generated by sweeping q'' from 2 W/cm^2 to the configuration CHF. The model contains no spatial PDE; it is a reduced-order network of algebraic closures, so the only numerical parameters are the root-finder tolerance, the flux-sweep resolution, and the discretization of the mechanistic coolant-side estimate.

3.5 Discretization sensitivity

Two numerical resolutions were verified (Figure 3). First, the mechanistic coolant-side estimate that motivates the resistance coefficient b integrates the local developing-laminar Nusselt number along each tube by a one-dimensional finite-volume marching scheme: the tube is divided into N axial segments, the coolant temperature is advanced segment by segment under an isothermal condensing wall, and the length-averaged internal Nusselt number is recovered from the log-mean temperature difference. The length-averaged $\overline{Nu}$ converges monotonically, changing by less than 0.4% between $N = 128$ and $N = 512$ for both chambers. Second, the predicted boiling curves and parity data

are sampled on a flux sweep; the curve is grid-independent to better than 0.05 K beyond 80 points. Production runs use 256 axial segments and 90 flux points, and the nonlinear surface-temperature solve converges to below 10^{-4} K, so all reported results are numerically converged.

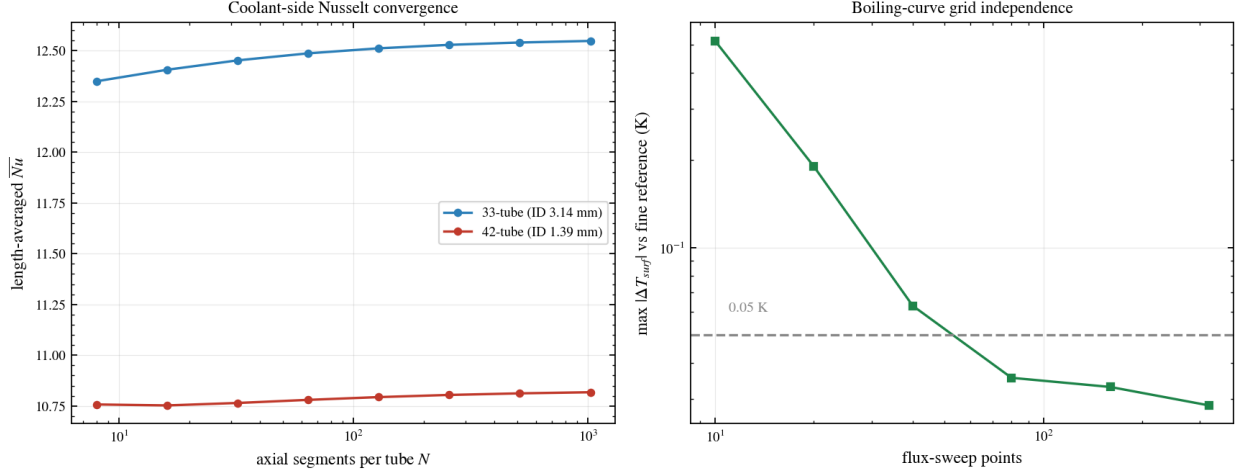


Figure 3: Discretization sensitivity. Left: length-averaged coolant-side Nusselt number versus axial segments per tube from the 1D marching scheme; both chambers plateau by a few hundred segments. Right: maximum deviation of the predicted boiling curve from a fine reference versus flux-sweep resolution.

3.6 Coefficient determination and justification

The calibrated constants are listed in Table 3; how each was obtained and why it is defensible follows.

Surface–fluid coefficients C_{sf} . Fit by least squares of Eq. (1) to the measured wall-superheat–flux data on each chip, all coolant inlets pooled. Both calibrated values land on tabulated Rohsenow coefficients for the matching copper finish (Table 2). The plain value 0.0131 agrees with water on polished copper, 0.0128–0.0130 [9, 10], to within 2%. The microchannel value 0.0067, roughly half, coincides with the classic value for water on *scored* copper, 0.0068 [10], which is physically apt: the open microchannels are an aggressive grooving of the surface that raises the active nucleation-site density [21]. That two coefficients fit only to this chamber’s data reproduce independent handbook values for the corresponding surface finish is direct evidence they are genuine surface–fluid properties rather than tuning constants. Section 4.2 adds a cross-chamber transfer test, and the identical plain boiling curve on both chambers confirms chamber independence.

Table 2: Calibrated C_{sf} against tabulated Rohsenow values for water on copper, ordered by value to pair each chip with its matching finish. Handbook values from [9, 10].

Surface (water on copper)	C_{sf}
Scored copper (handbook)	0.0068
Microchannel chip (this work)	0.0067
Polished copper (handbook)	0.0128–0.0130
Plain chip (this work)	0.0131
Lapped copper (handbook)	0.0147

Area augmentation $A_r = 2.17$. Not fitted. It is the geometric wetted-to-footprint area ratio of the Cooke–Kandlikar microchannel, computed from the channel and fin dimensions. Using the geometric value rather than a free multiplier preserves the predictive intent.

CHF increment $\Delta q''_{CHF} = +60 \text{ W/cm}^2$. Obtained from the measured plain-to-microchannel CHF differences, which are +64 and +59 W/cm^2 on the two chambers. The additive constant fits both far better than multiplying by A_r , which would over-predict the microchannel CHF to ~ 247 and $\sim 141 \text{ W/cm}^2$. A roughly fixed flux increment is consistent with a rewetting-limited enhancement that is largely independent of bundle geometry.

Single-phase factor F_{nc} . Fit per chamber on the low-flux, pre-boiling data to match the single-phase slope. It lumps subcooled-boiling onset and bulk circulation and is larger for the open 33-tube (9.0) than the confined 42-tube (3.5), consistent with stronger circulation in the open layout. It is chamber-specific, not transferable.

Operating-point coefficients (a, b, c). Per-chamber multilinear regression of measured T_{sat} on coolant inlet and chamber heat load, pooling plain and microchannel points across all coolant inlets. The coolant gain a is below unity (0.26 open, 0.71 confined), expressing the coolant-side limitation and the conductance improvement as the chamber warms; the slope b is an effective resistance whose magnitude is consistent with the 1D developing-laminar estimate of Figure 3; c is an offset. The regressions give a saturation-temperature RMSE of about 4 K (33-tube) and 9 K (42-tube).

Subcooling coefficients (d, e, f). Per-chamber regression of measured subcooling on coolant inlet and heat load. The 33-tube fit is well determined ($R^2 \approx 0.81$); the 42-tube fit is essentially scatter ($R^2 \approx 0.06$) and is flagged as a limitation.

Table 3: Calibrated model constants. C_{sf} and A_r are transferable; F_{nc} and the operating-point coefficients are per chamber. b and e are in K/W.

Group	Constant	33-tube	42-tube
Evaporator	C_{sf} (plain)	0.0131	
	C_{sf} (microchannel)	0.0067	
	Area augmentation A_r (microchannel)	2.17	
	Single-phase factor F_{nc}	9.0	3.5
CHF	q''_{CHF} plain (W/cm^2)	114	65
	Microchannel increment $\Delta q''_{CHF}$ (W/cm^2)	+60	
Operating T_{sat}	a (coolant gain)	0.262	0.710
	b (K/W)	5.99×10^{-3}	2.41×10^{-2}
	c (K)	44.8	33.1
Subcooling	d	−0.639	−0.217
	e (K/W)	$−6.80 \times 10^{-3}$	$−7.01 \times 10^{-4}$
	f (K)	40.8	19.6

3.7 Microchannel coefficient identifiability

Because the open-microchannel chip is the same physical surface in both chambers, its surface coefficient should be a single transferable value, and refitting it tests that. Fit chip-side against the measured operating point, the microchannel C_{sf} minimizes at 0.0070 on the 33-tube and 0.0064

on the 42-tube (Fig. 4a), bracketing the adopted pooled value 0.0067; the two chambers agree to within the width of a shallow minimum, and refitting C_{sf} alone changes the chip-side RMSE by less than 0.01 K. The coefficient is therefore well determined and chamber-independent, consistent with the plain-chip transfer test of Section 4.2.

Freeing the area augmentation A_r jointly with C_{sf} does not help. The two parameters both multiply the boiling heat transfer, so they trade off along a degenerate valley (Fig. 4b): pooled, the minimum drifts to (0.0084, 3.03) for a chip-side gain of only 0.4 K, while the per-chamber joint fits split to (0.0064, 1.93) and (0.0091, 3.50) and, applied end to end, trade one chamber for the other, improving the 42-tube microchannel RMSE from 5.2 to 4.5 K but worsening the 33-tube from 4.8 to 5.2 K. Holding A_r at its geometric value 2.17 and C_{sf} at the pooled 0.0067 is thus the parsimonious choice: it keeps the area a measured quantity rather than a fitted one and avoids overfitting a two-parameter enhancement to limited data. The residual microchannel error is not a deficiency of the surface coefficient; with the operating point taken from the data, the chip-side RMSE (5.6 and 7.1 K) exceeds the end-to-end RMSE (4.8 and 5.2 K), which means the smooth operating-point map averages out scatter that the raw measured saturation and subcooling carry, and that scatter, not the chip coefficient, sets the floor.

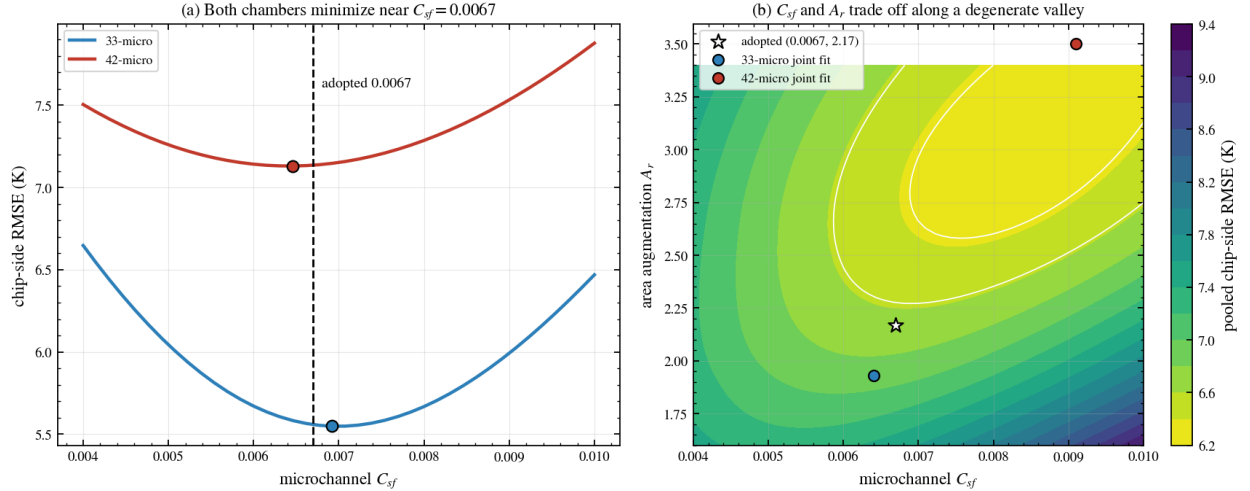


Figure 4: Microchannel coefficient identifiability. (a) Chip-side RMSE versus microchannel C_{sf} with the area augmentation fixed: both chambers minimize near the adopted 0.0067. (b) Pooled chip-side RMSE over (C_{sf}, A_r) : the two parameters trade off along a degenerate valley, and the per-chamber joint fits (markers) split apart, so the single geometric area plus the pooled coefficient is the robust choice.

3.8 Parameter sensitivity

Figure 5 shows how far the predicted chip temperature moves when each calibrated constant is perturbed by $\pm 10\%$, averaged over all points. The ranking is informative. The operating-point offset c dominates (3.2 K) and the coolant gain a is second (1.5 K), because both shift the predicted saturation temperature, which propagates almost directly into chip temperature. The boiling and single-phase constants are secondary: a $\pm 10\%$ change in C_{sf} moves the chip temperature under 1 K and A_r under 0.3 K. Two conclusions follow. The model is not hypersensitive to any single chip constant, so the few-percent uncertainty in C_{sf} or A_r established above is not limiting. And the accuracy is governed mainly by the empirical operating-point map, the same component the out-of-sample test (Section 4.2) identifies as the weakest link and that a direct condenser closure

would most improve.

Parameter sensitivity

Effect on chip temperature of a +/-10% change in each calibrated constant.

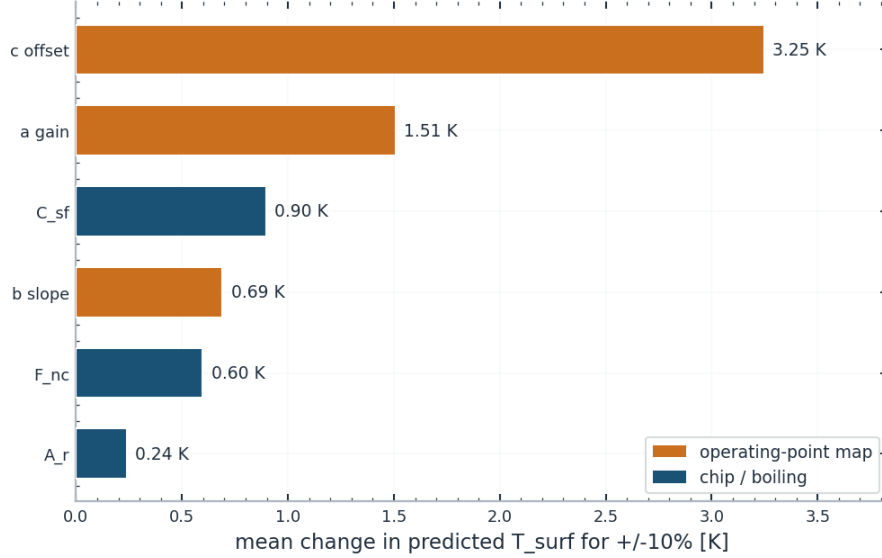


Figure 5: Sensitivity of predicted chip temperature to a $\pm 10\%$ change in each calibrated constant, averaged over all 261 points. Operating-point constants (orange) dominate; chip constants (blue) are secondary.

3.9 Measurement uncertainty

The error metrics are referenced to the measured chip temperature, which carries its own experimental uncertainty. Following the laboratory’s established procedure [5, 6], uncertainties are propagated by the root-sum-square (Kline–McClintock) method, with contributions from thermocouple calibration, the copper thermal conductivity, the thermocouple spacing, and the instrument least count. The chip surface temperature is obtained by extrapolating the embedded thermocouple readings to the boiling surface through the copper conduction length:

$$T_{surf} = T_{tc} - q'' \frac{\Delta x}{k_{Cu}}, \quad U_{T_{surf}}^2 = U_{T_{tc}}^2 + \left(\frac{\Delta x}{k_{Cu}} \right)^2 U_{q''}^2 + \left(\frac{q''}{k_{Cu}} \right)^2 U_{\Delta x}^2. \quad (9)$$

The calibrated thermocouple uncertainty is $U_{T_{tc}} = \pm 0.05$ K, the sensor-to-surface spacing $\Delta x = 5$ mm at $U_{\Delta x} = \pm 0.01$ mm, and the copper conductivity $k_{Cu} = 401 \pm 9$ W m⁻¹K⁻¹ (2.2%). The heat flux follows from the one-dimensional gradient $q'' = -k_{Cu} (dT/dx)$ fit to the thermocouple column; its relative uncertainty is dominated by the random scatter in that gradient at low flux and by the systematic (k_{Cu} , spacing) floor at high flux. For the present data it falls from 10.9% at the lowest flux (~ 9 W cm⁻²) to a systematic floor of about 2.5% (set by k_{Cu} and the spacing) at high flux. The boiling curves are logged continuously over the full nucleate range up to the critical heat flux of each configuration, reaching 174 W cm⁻² on the microchannel 33-tube (Table 4); the high-flux nucleate branch is therefore measured, not a model extrapolation anchored only at a separate CHF point, and the reported critical heat fluxes carry about $\pm 4\%$. The conduction correction $q'' \Delta x / k_{Cu}$ grows with flux, reaching ~ 22 K at 174 W cm⁻², so its contribution to $U_{T_{surf}}$ scales with load: $U_{T_{surf}}$ is within ± 0.2 K at low and moderate flux and widens to about ± 0.5 K at

the CHF points. The saturation temperature is set by the chamber pressure through the vapor-pressure slope, $U_{T_{sat}} = (dT_{sat}/dP) U_P$ with $dT_{sat}/dP \approx 0.28$ K/kPa for water and HFE-7000 near their saturation, so a transducer uncertainty of order 0.5 kPa gives $U_{T_{sat}} \approx \pm 0.15$ K and a wall-superheat uncertainty within ± 0.25 K. The coolant-side balance $Q = \dot{m} c_p (T_{out} - T_{in})$ provides an independent check on the chip flux. These bands set the noise floor against which the model error should be read: even at the worst-case ± 0.5 K surface-temperature uncertainty reached at CHF, the chip-temperature RMSE of 4.4 K exceeds it by roughly an order of magnitude, so the residual reflects model structure, chiefly the operating-point regression, rather than measurement noise.

4 Results and validation

4.1 Model versus experiment for each chip and condenser

Figure 6 compares the fully predictive model against experiment for each chip and condenser separately, plotting heat flux against chip surface temperature with the model swept to CHF. Per-configuration errors are in Table 4; the aggregate parity is Figure 7.

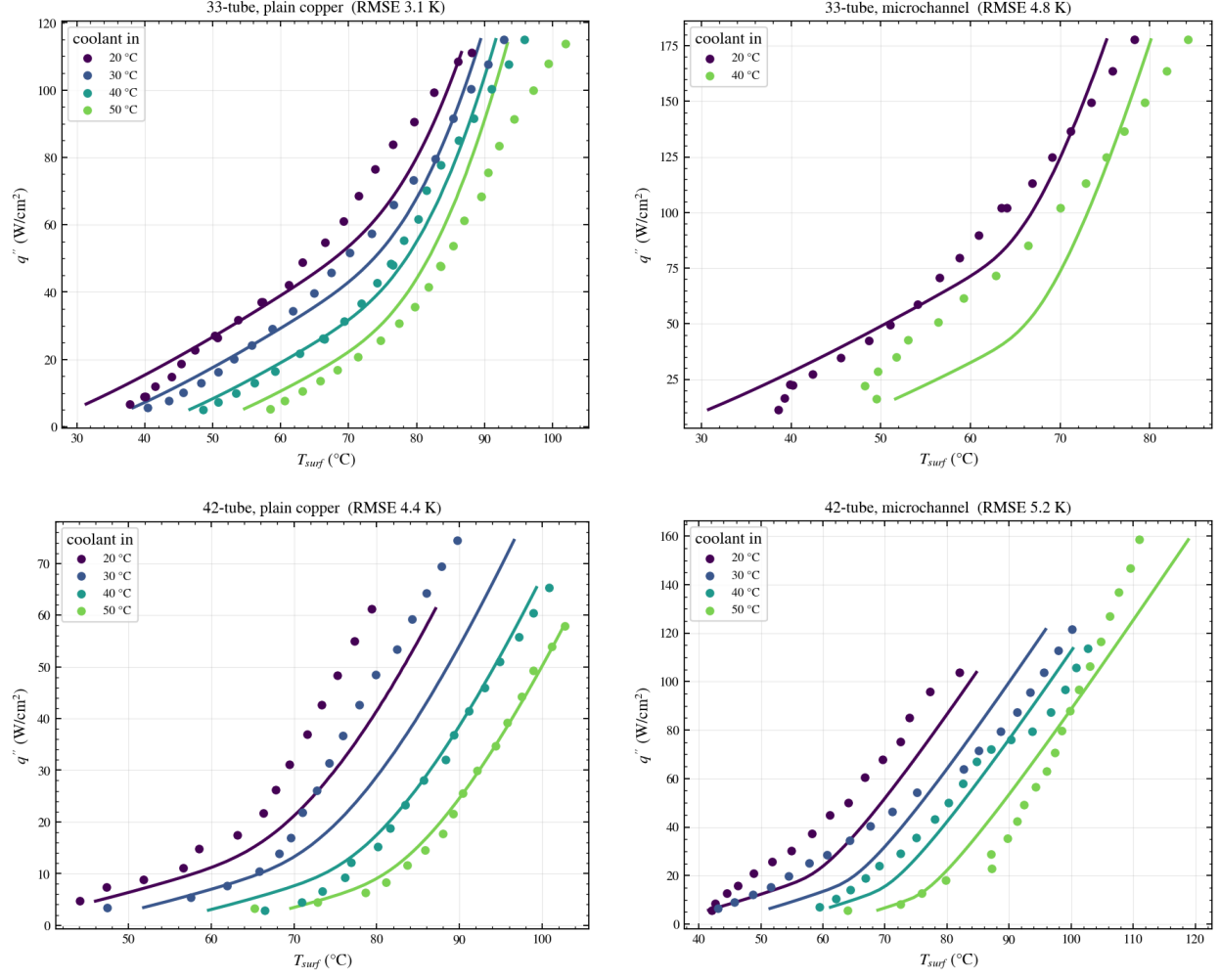


Figure 6: Heat flux versus chip surface temperature for each chip and condenser, model swept to CHF. Markers are measured (distinct symbol per coolant inlet); lines are the fully predictive model.

Table 4: Chip-temperature prediction error (fully predictive).

Configuration	RMSE (K)	CHF (W/cm ²)
33-tube, plain	3.2	114
42-tube, plain	4.5	65
33-tube, microchannel	4.9	174
42-tube, microchannel	5.2	125
All points	4.4	—

End-to-end predictive accuracy

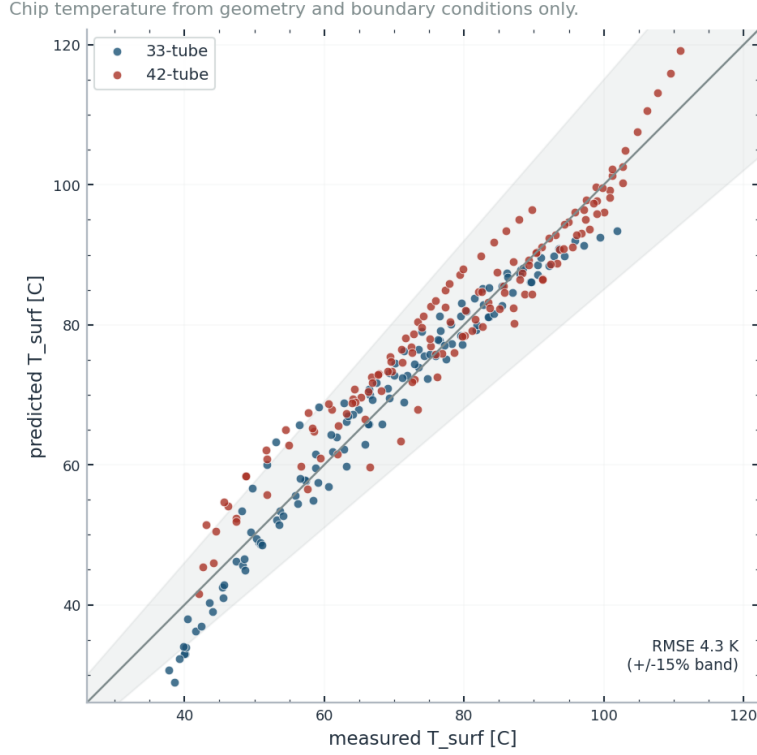


Figure 7: End-to-end parity of predicted versus measured chip surface temperature. The shaded band is $\pm 15\%$.

4.2 Out-of-sample validation

The errors above are computed on the same points used for calibration. To test the central claim that the model predicts rather than fits, two out-of-sample exercises were run (Figure 8, Table 5). First, a leave-one-coolant-inlet-out test: for each chamber the operating-point and subcooling maps were refit on three of the four coolant inlets and used to predict chip temperature at the withheld inlet, with the mechanistic chip model held fixed. The 33-tube generalizes well, the held-out RMSE rising only from 3.7 to 4.9 K, and the interior inlets (30 and 40 °C) are predicted at in-sample quality; the edge inlets (20 and 50 °C), which require extrapolating the linear map, are the hardest. The 42-tube degrades more, from 4.8 to 9.1 K, consistent with its higher coolant gain (0.71 versus 0.26), which makes the operating point depend strongly on coolant inlet, and with its poorly determined subcooling fit.

Second, a cross-chamber transfer of the plain-chip surface coefficient. Fit independently on the two geometrically different chambers, C_{sf} comes out at 0.0130 (33-tube) and 0.0121 (42-tube), nearly identical and both close to the pooled value 0.0131. Applying the coefficient fit on one chamber to the other raises the plain-chip RMSE by at most 0.2 K (33-tube 3.2 to 3.4 K; 42-tube 4.4 to 4.5 K). This directly supports treating C_{sf} as a transferable surface-fluid property rather than a per-chamber fit, which is the basis for predicting an untested chamber.

Out-of-sample validation

Chip temperature predicted for a coolant inlet withheld from the fit.

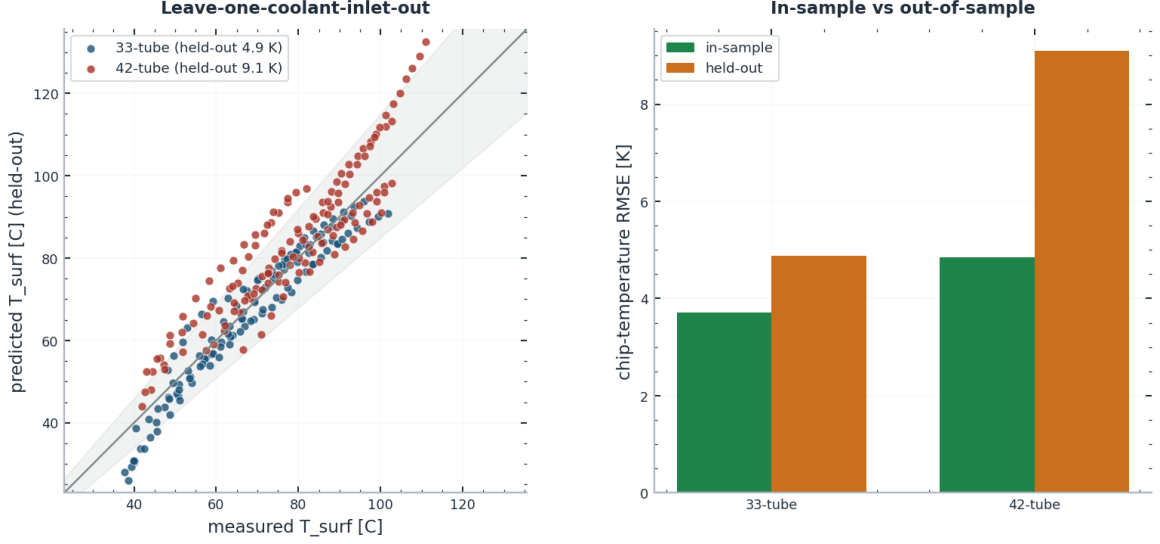


Figure 8: Out-of-sample validation. Left: chip temperature predicted for a coolant inlet withheld from the operating-point fit (held-out points only). Right: in-sample versus held-out error, with the plain-chip C_{sf} recovered independently on each chamber.

Table 5: Out-of-sample validation. Leave-one-coolant-inlet-out (LOCO) refits the operating-point and subcooling maps on three coolant inlets and predicts the fourth; the transfer test applies a plain-chip C_{sf} fit on one chamber to the other.

	33-tube	42-tube
In-sample chip-temperature RMSE (K)	3.7	4.8
LOCO held-out chip-temperature RMSE (K)	4.9	9.1
LOCO held-out T_{sat} RMSE (K)	7.4	15.3
C_{sf} fit on this chamber alone	0.0130	0.0121
Plain-chip RMSE, native C_{sf} (K)	3.2	4.4
Plain-chip RMSE, transferred C_{sf} (K)	3.4	4.6

4.3 Origin of the 42-tube generalization gap

The leave-one-coolant-out gap between the chambers, 4.9 K held-out RMSE for the 33-tube against 9.1 K for the 42-tube, is worth isolating because both chambers share the same mechanistic chip model; only the operating-point map differs. Figure 9 traces the gap to its source, and the chip physics is not it. Holding the true operating point fixed, the two chambers predict surface temperature at the same quality; the difference lives entirely in how well the chamber pressure can be predicted.

For the open 33-tube bundle the condenser-side resistance $(T_{liq} - T_{in})/Q$ is nearly flux-independent, a median near 13 mK/W across the full flux range (Fig. 9a). For the confined 42-tube bundle it is strongly flux-dependent, falling from roughly 150 mK/W at low flux to about 45 mK/W near CHF, a slope of -2.3 mK/W per W/cm^2 against the 33-tube's -0.07 . The confined bundle is

coolant-starved at low flux and progressively better swept as boiling intensifies, so its pool-to-coolant resistance, and therefore the chamber pressure and saturation temperature it sets, depend nonlinearly on heat load. The linear operating-point map $T_{sat} = aT_{in} + bQ + c$ cannot represent this: it fits the 42-tube saturation temperature with an 8.6 K RMS residual against 4.1 K for the 33-tube.

That operating-point error transmits almost one-to-one to surface temperature. Perturbing the saturation temperature in the chip model gives $dT_{surf}/dT_{sat} \approx 0.9$ for both chambers, since in nucleate boiling the wall tracks saturation. Figure 9b confirms it: across the held-out points the surface-temperature error follows the saturation-temperature error along the 0.9 line, and the 42-tube simply spans a far wider band of operating-point error, about ± 35 K against ± 15 K. The three metrics line up accordingly (Fig. 9c): the 42-tube is worse by the same factor of roughly two on the saturation-map residual (8.6 versus 4.1 K), the held-out saturation error (15.3 versus 7.4 K), and the held-out surface-temperature RMSE (9.1 versus 4.9 K). The gap is thus a limitation of representing a flux-dependent confined-bundle condenser with a load-linear map, not of the boiling model. The implication for an untested confined geometry is concrete: the operating-point sub-model should carry a mechanistic flux-dependent condenser resistance rather than a linear-in- Q fit, which is the single change most likely to bring 42-tube-class generalization in line with the 33-tube.

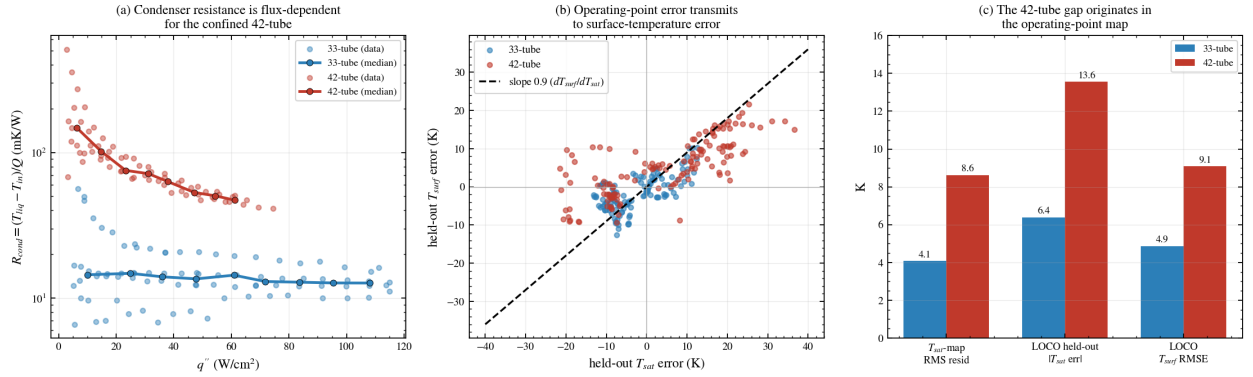


Figure 9: Origin of the 42-tube generalization gap. (a) Condenser-side resistance versus heat flux: flat for the open 33-tube, strongly flux-dependent for the confined 42-tube (binned medians over the data cloud). (b) Held-out surface-temperature error versus held-out saturation-temperature error; both follow the $dT_{surf}/dT_{sat} \approx 0.9$ line, the 42-tube over a wider range. (c) The 42-tube is worse by about a factor of two on the saturation-map residual, the held-out saturation error, and the held-out surface-temperature RMSE alike.

4.4 Comparison with baseline models

To show that the two main modeling choices earn their place, the full gray-box is compared against two reduced alternatives (Table 6). The first replaces the per-surface boiling coefficients with a single literature value $C_{sf} = 0.013$ for both chips. On the plain chip this is indistinguishable from the calibrated model (3.8 versus 3.8 K), confirming that the plain calibration only recovers the handbook value; on the microchannel it nearly doubles the error (8.8 versus 5.1 K), because the structured surface genuinely needs its own lower coefficient. The second replaces the calibrated operating-point map with an ideal effectiveness–NTU condenser using fully developed laminar coolant Nusselt number and the nominal tube area. Its effectiveness is only 0.03–0.04, so it over-predicts the saturation temperature by a median of 30 K and drives a few points above the critical point, which is nonphys-

ical. Both comparisons reproduce the paper’s central claims: the microchannel requires a distinct surface coefficient, and the condenser must be modeled as coolant-side limited rather than by ideal effectiveness–NTU.

Table 6: Baseline comparison. Chip-temperature RMSE (K) for the full model versus a single- C_{sf} evaporator; the effectiveness–NTU condenser is reported by its saturation-temperature error, being nonphysical at high flux.

Model	All	Plain	Microchannel
Full gray-box	4.4	3.8	5.1
Single $C_{sf} = 0.013$, both chips	6.4	3.8	8.8
Effectiveness–NTU condenser: effectiveness 0.03–0.04; median T_{sat} over-prediction 30 K; 3/261 points nonphysical ($> T_{crit}$).			

4.5 Condenser operating behavior

Figure 10 shows how the condenser sets the operating point as the two control parameters, coolant inlet temperature and chamber heat load, are varied. Removing the load contribution with the fitted map isolates the coolant-inlet sensitivity (top left): saturation temperature rises with coolant inlet at the coolant gain a , which is 0.26 for the open 33-tube and 0.71 for the confined 42-tube, so the open chamber largely pins its saturation temperature while the confined one tracks the coolant. Removing the inlet contribution isolates the load sensitivity (top right), whose slope is the effective resistance b , again much steeper for the 42-tube. Pool subcooling falls with heat load (bottom left), and the condenser-side resistance (bottom right) is roughly flux-independent at ~ 13 mK/W for the 33-tube and ~ 30 – 45 mK/W for the 42-tube, consistent with the breakdown below.

Condenser operating behavior

Saturation temperature and subcooling vs coolant inlet and heat load.

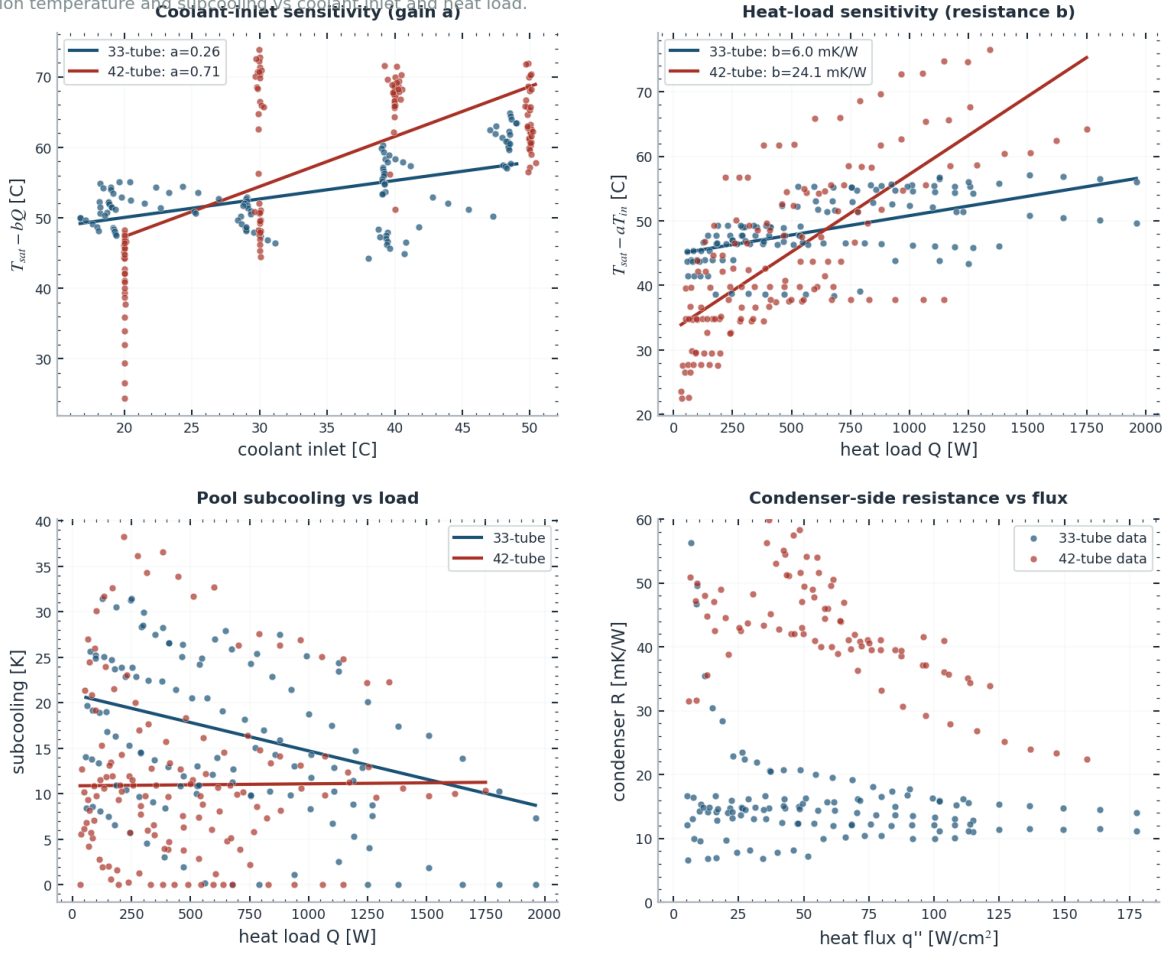


Figure 10: Condenser operating behavior. Saturation temperature versus coolant inlet (coolant gain) and heat load (effective resistance), pool subcooling versus load, and condenser-side resistance versus flux. Markers are data pooled over both chips; lines are the operating-point model.

4.6 Thermal resistance breakdown

Figure 11 gives the chip-side and condenser-side resistances versus heat flux to CHF for each configuration, referenced to the bulk liquid temperature; values near CHF are in Table 7. Referenced to saturation, the boiling heat transfer coefficient is ≈ 35 kW/m²K for the plain chip and up to 200 kW/m²K for the microchannel.

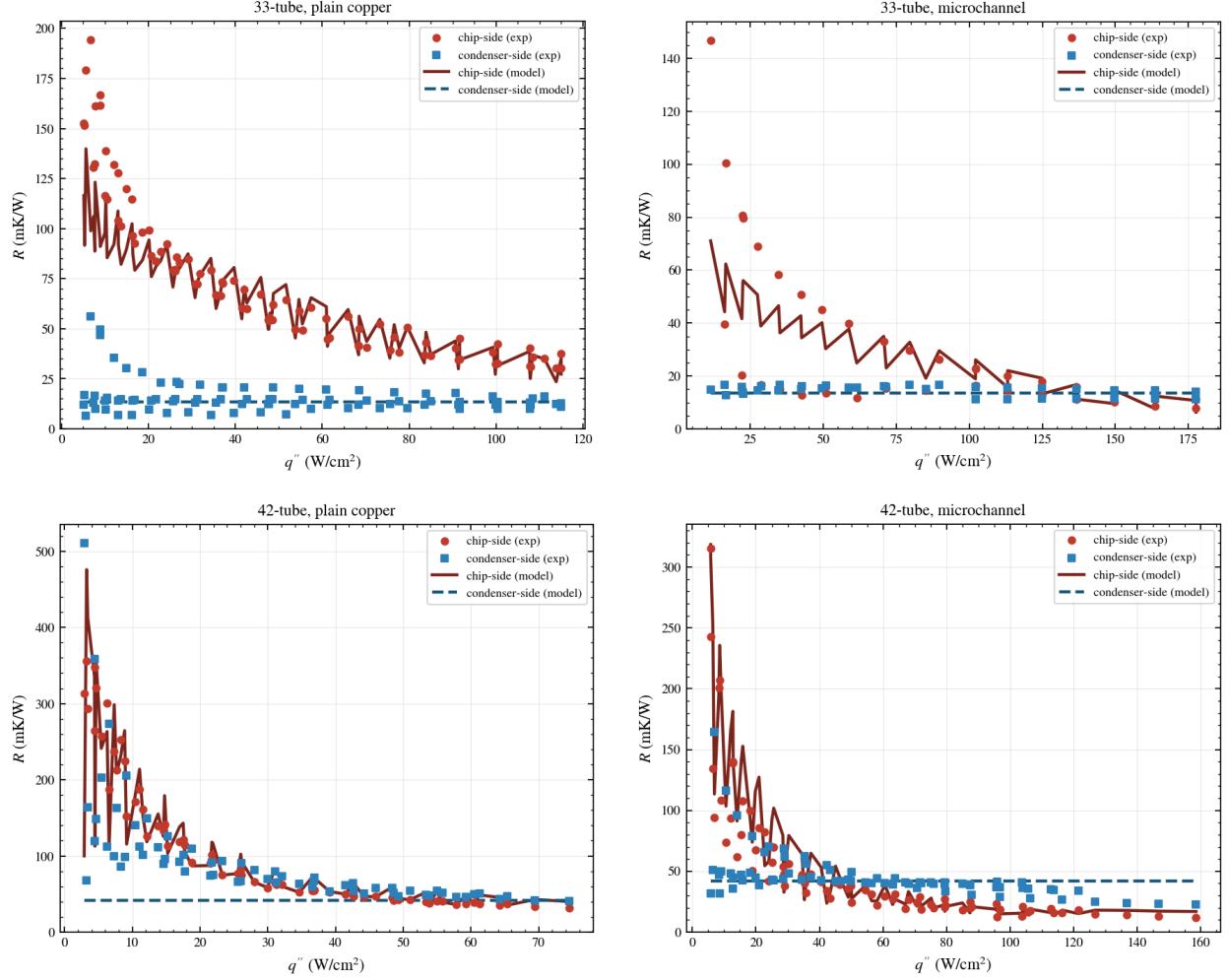


Figure 11: Chip-side and condenser-side thermal resistance versus heat flux to CHF, referenced to bulk liquid temperature. Markers are data (distinct symbol per side); lines are the model.

Table 7: Thermal resistance near CHF, referenced to bulk liquid, $R = \Delta T/Q$ in mK/W.

Configuration	R_{chip}	R_{cond}	R_{tot}
33-tube, plain	33	13	47
33-tube, microchannel	11	13	24
42-tube, plain	36	46	81
42-tube, microchannel	15	26	41

The chip-side resistance is compared directly against experiment for both surfaces in Figure 12. The microchannel's roughly threefold reduction is evident, from ~ 38 to ~ 13 mK/W near CHF on the 33-tube, and the model tracks the experimental chip-side resistance for both chips, the largest deviation being the 42-tube plain configuration, consistent with its higher overall error.

Chip-side thermal resistance: model vs experiment

Microchannel cuts chip-side resistance about threefold; model tracks both surfaces.

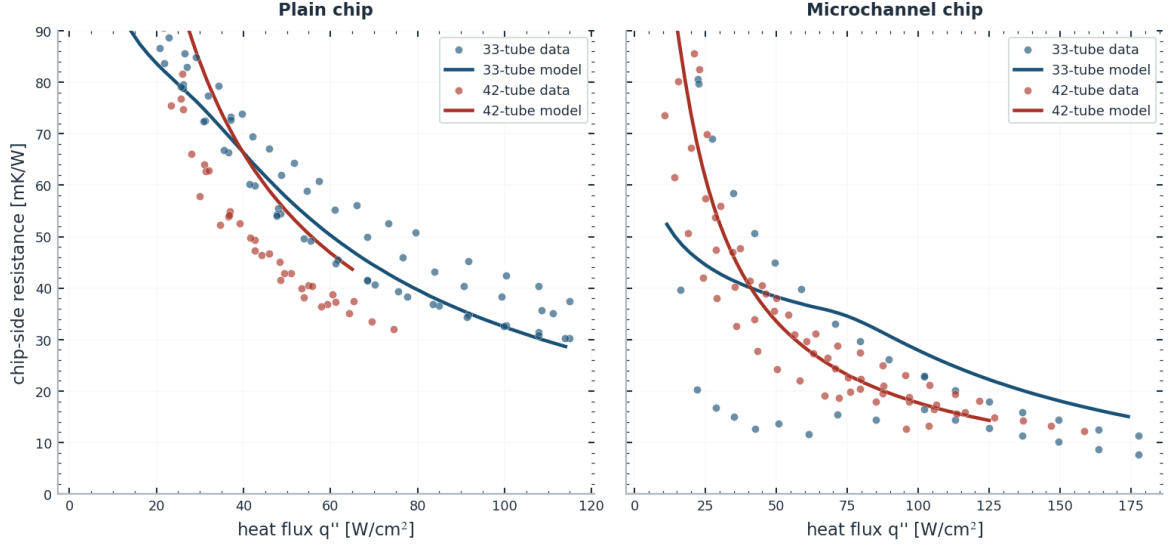


Figure 12: Chip-side thermal resistance $R_{chip} = (T_{surf} - T_{bulk})/Q$ versus heat flux, model versus experiment, referenced to bulk liquid. The microchannel cuts the chip-side resistance about threefold relative to the plain chip; the model reproduces both surfaces.

5 Discussion

The central physical finding is that the condenser is limited by coolant-side developing-laminar convection rather than by condensation, and that its conductance improves with operating temperature. This produces a saturation-temperature gain on coolant inlet below unity and explains why the open 33-tube self-regulates while the confined 42-tube, with smaller, denser, more submerged tubes, climbs. The resistance split referenced to bulk liquid shows that the chip side, which carries the pool subcooling as well as the wall superheat, is the larger resistance for the open 33-tube, whereas the condenser side dominates for the confined 42-tube. The microchannel cuts the chip-side resistance roughly threefold, after which the condenser side is comparable (33-tube) or dominant (42-tube). The design consequence is explicit: surface enhancement of the chip is effective, but for the resistive 42-tube the condenser is the bottleneck, and further gains require improving it rather than the chip. The model also separates transferable quantities (C_{sf} , A_r) from chamber-specific ones (F_{nc} and the operating-point coefficients), the latter being physically interpretable.

5.1 Predictive scope for untested geometry

The two sides of the model are predictive to different degrees, which determines what can be forecast for hardware that has not been tested.

New chip or area geometry: predictive. The evaporator is fully mechanistic. The surface coefficient C_{sf} is a transferable surface–fluid property (Sections 3.6, 4.2): it transfers across chambers at under 0.2 K and matches handbook values by surface finish. The area augmentation A_r is computed from the channel geometry, not fitted. A new boiling surface therefore needs only its C_{sf} , assigned from its finish or pinned by a single boiling curve, together with its geometric A_r ; the

full $T_{surf}(q'')$ curve then follows with no chamber-specific evaporator fitting. The exception is the burnout limit: the microchannel CHF is an additive *measured* increment and the mechanistic CHF model is an uncalibrated placeholder, so the curve is predictable up to CHF but the CHF value itself needs a measurement or a dedicated closure.

New condenser geometry: ranking predictable, absolute level needs one calibration. The coolant-side resistance follows from geometry through the developing-laminar internal-flow correlation, and it sets the right order of magnitude and ordering (Table 8, Figure 13): the measured condenser-side resistance is 13.5 mK/W for the 33-tube against a geometric estimate of 17.3, and 42 mK/W for the 42-tube against 25.7. The two chambers bracket the ideal, the open 33-tube does slightly better than full-area developing-laminar convection (additional shell-side mechanisms), the confined 42-tube does worse (only part of the bundle actively condenses), so the ratio of real to ideal resistance is a layout-dependent factor (0.8 open, 1.6 confined) that tube dimensions alone do not pin. As a result the absolute operating point (the T_{sat} and subcooling maps) is currently fit per condenser from one dataset; once fit, the model predicts across coolant inlets, fluxes, and chips for that condenser. Making the condenser side fully geometry-predictive, replacing the operating-point map with an active-area closure that returns (a, b, c) from geometry and fill, is the main open step, and it requires at least a third characterized condenser to fit and test.

The 66-tube. Its coolant-side resistance is the lowest of the three by geometry (~ 11 mK/W at the 33-tube coolant flow). The tubes are the same 95 mm length as the 33-tube and the external area is in fact smaller (394 versus 469 cm², Table 9); the lower resistance comes from the smaller bore (1.80 mm ID versus 3.14 mm), which raises the internal developing-laminar convective coefficient enough to more than offset the reduced external area. On condenser hardware alone it should therefore hold the lowest T_{sat} for a given load and run the coolest chip. Being open like the 33-tube, its real-to-ideal factor is expected near the 33-tube’s, which bounds its operating point close to the geometric estimate. This remains a prediction until one 66-tube dataset is taken, which is the recommended next measurement.

Table 8: Condenser-side thermal resistance: geometric developing-laminar coolant-side estimate versus measured. The real-to-ideal ratio is layout-dependent, so it is not fixed by tube dimensions alone.

Condenser	Re	Geometric R (mK/W)	Measured R (mK/W)	real/ideal
33-tube (open)	1200	17.3	13.5	0.78
42-tube (confined)	985	25.7	42.0	1.63
66-tube (open)	1050	11.1	—	—

Condenser resistance: geometry vs measurement

Developing-laminar estimate sets the ranking; a layout factor separates ideal from real.

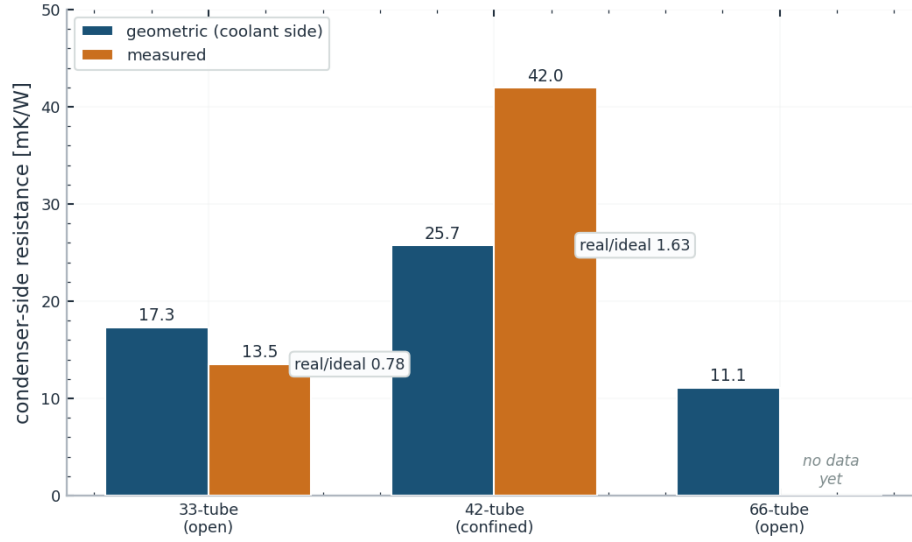


Figure 13: Condenser-side resistance from geometry versus measurement. The developing-laminar coolant-side estimate sets the ranking and order of magnitude; a layout-dependent factor separates ideal from real. The 66-tube has the lowest geometric resistance but is not yet measured.

5.2 Predicted 66-tube performance

The 66-tube operating point is not measured, but its one geometric difference, a lower condenser-side resistance, can be propagated through a construction that is first validated on the 33-tube. The pool temperature is set by the condenser, $T_{bulk} = T_{in} + Q R_{cond}$, with R_{cond} the developing-laminar coolant-side estimate scaled by the open real-to-ideal factor; the saturation temperature is $T_{sat} = T_{bulk} + \Delta T_{sub}$ with subcooling from the open-bundle map; and the chip temperature follows from the transferable boiling model. Applied to the 33-tube this reproduces the calibrated model to within 0.2 K, so the same construction is used for the 66-tube with its geometric resistance of 8.7 mK/W (11.1×0.78).

Tables 9 and 10 give the geometry and the predicted operating point at CHF. Rejecting heat through roughly half the condenser-side resistance of the 33-tube, the 66-tube holds a cooler pool and a lower saturation temperature for the same load: the chip runs about 4 K cooler on the plain surface and 8 K cooler on the microchannel at CHF, the ~ 6 K drop in saturation temperature outweighing a small rise in wall superheat at the lower pressure. The condenser-side resistance falls from 0.0135 to 0.0087 K/W while the chip-side resistance, fixed by the same surface and flux, is nearly unchanged. These are predictions under stated assumptions, CHF and pool subcooling taken from the open 33-tube, that one 66-tube dataset would replace and test directly.

Table 9: Geometry of the three condensers. Internal and external areas are $\pi (ID) L N$ and $\pi (OD) L N$. The 66-tube coolant flow is assumed equal to the 33-tube loop (*).

	33-tube	42-tube	66-tube
Tubes N	33	42	66
Outer diameter (mm)	4.76	3.18	2.00
Inner diameter (mm)	3.14	1.39	1.80
Tube length (mm)	95	62.6	95
Internal area (cm ²)	309	115	355
External area (cm ²)	469	262	394
Coolant flow (g/s)	78	36	78*
Coolant Re	1200	985	1050
Manifolds	open ext.	in-chamber	open ext.
Geometric coolant-side R (mK/W)	17.3	25.7	11.1

Table 10: Predicted 66-tube performance at CHF (coolant inlet 30 °C), with the 33-tube for comparison. CHF is taken equal to the 33-tube; the resistance split is referenced to bulk liquid.

	66-tube (predicted)		33-tube	
	plain	microch.	plain	microch.
q''_{CHF} (W/cm ²)	114	174	114	174
Heat load Q (W)	1259	1921	1259	1921
T_{sat} (°C)	53.9	55.2	60.0	64.5
Subcooling (K)	13.0	8.5	13.0	8.5
T_{surf} at CHF (°C)	85.6	69.8	89.4	77.5
R_{chip} (K/W)	0.0355	0.0121	0.0337	0.0112
R_{cond} (K/W)	0.0087	0.0087	0.0135	0.0135
R_{tot} (K/W)	0.0442	0.0207	0.0472	0.0247

Figure 14 shows the predicted boiling curve obtained by sweeping the same construction to CHF. The left panel is the boiling curve proper, heat flux against wall superheat: below boiling onset the surface sits in subcooled single-phase convection, and above it the nucleate branch rises steeply, with the microchannel reaching 174 W/cm² at about 15 K superheat against 32 K for the plain chip. The right panel plots flux against chip temperature with the 33-tube dashed for reference, showing the 66-tube holding a lower chip temperature at every flux because of its lower condenser-side resistance.

Predicted boiling curve for the 66-tube condenser

Coolant inlet 30 °C. CHF taken from the 33-tube pending a 66-tube measurement.

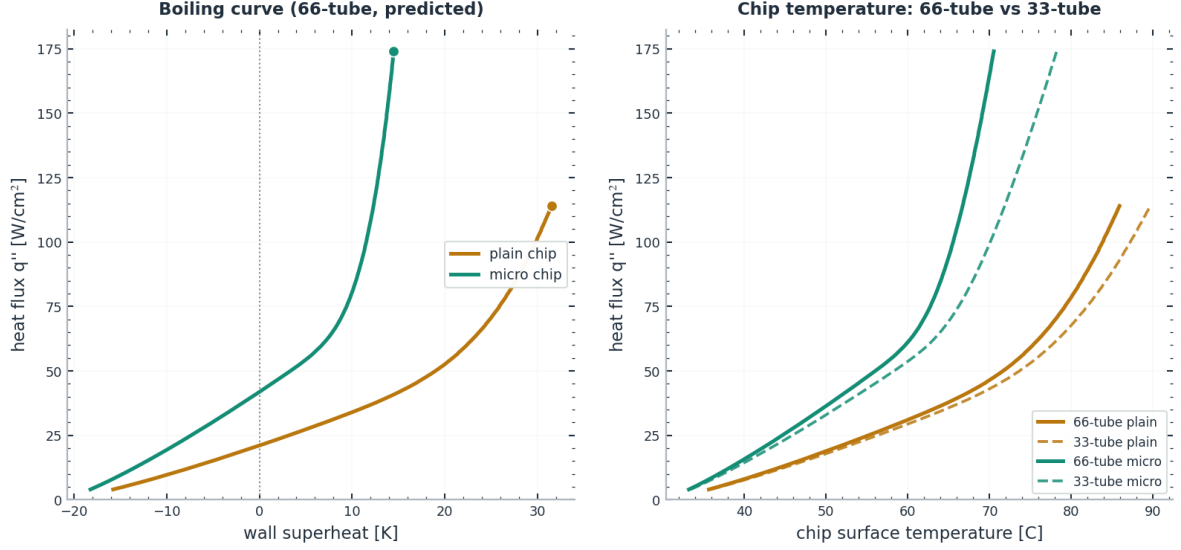


Figure 14: Predicted 66-tube boiling curve at 30 °C coolant inlet. Left: heat flux versus wall superheat (single-phase below onset, nucleate above) to CHF. Right: heat flux versus chip surface temperature, with the 33-tube (dashed) for comparison. Curves use the geometry-based construction validated against the 33-tube; CHF is taken from the 33-tube pending a 66-tube measurement.

6 Conclusions

1. A reduced-order model predicts chip surface temperature from heat flux, coolant inlet temperature, chip surface, and condenser geometry, with no measured operating conditions, to a root-mean-square error of 4.4 K (median 4% of absolute temperature; 5% on the rise above coolant in the high-flux regime).
2. The microchannel chip is captured by the same boiling machinery with a fixed area augmentation ($A_r = 2.17$) and a halved, transferable surface coefficient ($C_{sf} = 0.0067$ versus 0.0131); its CHF is the plain CHF plus a constant ~ 60 W/cm².
3. The condenser is coolant-side limited, and its coolant gain (0.26 open, 0.71 confined) is a fingerprint of the chamber.
4. Referenced to bulk liquid, the condenser side dominates the chip-to-coolant resistance for the confined 42-tube and is comparable to the chip side for the 33-tube once the microchannel is used, identifying the condenser as the target for further improvement.
5. Out-of-sample tests support the predictive claim: the plain-chip surface coefficient fit independently on the two chambers agrees (0.0130 versus 0.0121) and transfers between them at a cost below 0.2 K, while a leave-one-coolant-inlet-out test predicts chip temperature at a withheld inlet with 4.9 K RMSE on the 33-tube (9.1 K on the 42-tube, limited by its stronger coolant-inlet dependence).
6. The model is predictive on the evaporator side, a new chip surface is forecast from a transferable C_{sf} and a geometric area augmentation, up to a CHF limit that still needs measurement,

whereas the condenser side predicts the resistance ranking from geometry but requires one calibration dataset to fix the absolute operating point, which sets the agenda for a geometry-based condenser closure.

7 Limitations and future work

The operating-point coefficients are a calibrated per-chamber map, not a first-principles closure, partly because the coolant flow carries measurement uncertainty and the level dynamics that grow the condensing area are not resolved from geometry alone. Subcooling is well predicted on the 33-tube and essentially scatter on the 42-tube. CHF is anchored to measured burnout. The immediate extension is a single run on the 66-tube condenser to fix its operating-point coefficients. The HFE-7000 path in the model is at present a literature-anchored prediction, with smooth-copper boiling coefficients rather than a chamber-data fit, so extending the model to a validated second working fluid, together with a mechanistic CHF closure, are longer-term goals that require an HFE dataset.

8 Next actions

The following are ordered by how much each converts a standing assumption into validated physics.

1. **Take one 66-tube dataset.** A single sweep on the existing 66-tube condenser validates the prediction in Section 5.2, fixes its operating-point coefficients, and provides the third characterized condenser needed to begin a geometry-based condenser closure. This is the single highest-value measurement.
2. **Resolve whether CHF is burnout or a surface-temperature ceiling.** The same plain chip reaches 114 W/cm^2 on the 33-tube and 65 on the 42-tube at comparable subcooling and similar surface temperature, which suggests a system limit rather than a local boiling crisis. Settling this from the existing records determines whether the 66-tube and HFE CHF values should be inherited, raised (lower-resistance condenser reaches a ceiling at higher flux), or modeled mechanistically.
3. **Bring in HFE-7000.** When data is available, refit C_{sf} for HFE on copper (expected near scored-copper values, with a much lower CHF), recompute the boiling curve, and test whether the coolant-side-limited operating-point structure holds for a second fluid. The El-Genk correlations give an independent check on the boiling coefficient and the subcooling dependence.
4. **Add error bars to the data figures.** With the measurement-uncertainty section now in place (Section 3.9), draw the propagated bands on the boiling-curve and resistance figures so the model error can be read directly against the noise floor.
5. **Document the physical apparatus.** Add a labeled apparatus schematic, the full experimental test matrix, and a thermophysical property table for the working fluids.
6. **Develop the geometry-based condenser closure.** Replace the empirical operating-point map with an active-area model that returns the coefficients from condenser geometry and liquid fill, which is the step that makes the condenser side fully predictive. This requires the 66-tube data and, ideally, the fill-ratio sweeps.

7. **Strengthen the 42-tube subcooling closure**, currently essentially scatter, and add a mechanistic CHF model to remove the last measured anchor.

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